

Math 252, Fourth Examination-Review

Directions: Respond to each question by showing all work and fully justifying your solution. Your justification should include any well-known theorems, formulas, or mathematical properties used. **You may use an entire page for each problem, including both the front and back.** Only a **TI-30XII** scientific calculator may be used; **Casio calculators are not permitted.** If you use a calculator for a calculation, you must also show the expression or calculation entered to obtain your result. Answers submitted without sufficient supporting work or justification may receive little or no credit. Or, if you prefer the directions as a bulleted list (which many instructors find easier for students to

- (1) Let the curve C be given by $x = -3 \sin 2t$ and $y = 3 \cos 2t$ for $0 \leq t \leq 3\pi$
 - (a) By eliminating the parameter, t , find the equation of the circle (centered at the origin) in rectangular coordinates.
 - (b) Determine the direction in which it is traversed and how many revolutions traversed over the interval.
- (2) Let the curve C be given by $x = -2 + 3t$ and $y = 2 - 3t$ for all t .
 - (a) By eliminating the parameter, t , determine an equation relating x and y to each other.
 - (b) Sketch a graph of C indicating the direction.
- (3) Let the curve C be defined parametrically by $x = t^3 - 3t$ and $y = t^2 - 2$, for $-2 \leq t \leq 2$.
 - (a) Find an expression for $\frac{dy}{dx}$ in terms of t .
 - (b) Determine all points, (x, y) , on C at which the tangent line is horizontal, **AND** all points (x, y) on C at which the tangent line is vertical.
- (4) Let the curve C be defined parametrically by $x = t^3$ and $y = t^4$, for $-2 \leq t \leq 2$.
 - (a) Determine whether the curve C is smooth. Justify your answer.
 - (b) Show that the point $(0, 0)$ has both a horizontal tangent line and a vertical tangent line.
- (5) Let the curve C be given parametrically by $x = 1 - t^2$ and $y = 1 + t^3$, for $0 \leq t \leq 1$. Find the length, L , of C .
- (6) Let the curve C be given parametrically by $x = e^t \sin t$ and $y = e^t \cos t$, for $0 \leq t \leq \pi$. Find the length, L , of C .
- (7) Consider the rose $r = 2 \sin(3\theta)$.
 - (a) Sketch a graph of the rose.

- (b) Write an integral and evaluate it for the area lying inside one petal of the rose.
- $$\sin^2 x = \frac{1 - \cos 2x}{2}$$

(8) Consider the rose $r = 2 + 2 \sin \theta$.

(a) Sketch a graph of the cardioid.

(b) Write an integral and evaluate it for the area lying inside of the cardioid.

(9) Consider the rose $r = 1 + 2 \cos \theta$.

(a) Sketch a graph of the limaçon.

(b) Write an integral and evaluate it for the area of the outter loop.